



Roll No

National Institute of Technology Goa

Cuncolim, Salcete, South Goa - 403703.

Programme Name: B.Tech.

Mid Semester Examinations, September 2026.

Course Name: **Advanced Differential Equations and Complex Analysis**

Course Code: **MA200**

Date: 23/09/2026

Time: 10.00 AM - 11.30 AM

Duration: 1 Hour 30 Minutes

Max. Marks: 50

ANSWER ALL QUESTIONS

1. Using Frobenius method, solve: $9x(1-x)y'' - 12y' + 4y = 0$. [6]
2. Find the power series solution of the initial value problem $(x^2 - 1)y'' + 3xy' + xy = 0, y(2) = 4, y'(2) = 6$. [6]
3. Prove that the Bessel's function $J_\nu(x)$ and its derivative satisfy the recurrence relations [8]
 - (a) $[x^\nu J_\nu(x)]' = x^\nu J_{\nu-1}(x),$
 - (b) $[x^{-\nu} J_\nu(x)]' = -x^{-\nu} J_{\nu+1}(x).$
4. Find all the eigenvalues and eigenfunctions of the Sturm-Liouville problem $y'' + \lambda y = 0$, subject to the boundary conditions $y(0) + y'(0) = 0, y(\pi) + y'(\pi) = 0$. [10]
5. (a) Prove that $P_n(x) = \frac{1}{n!} \frac{d^n}{dx^n} (x^2 - 1)^n$. [5]
(b) Express $4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomials. [5]
6. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is vibrating by giving to each of its points a velocity $\lambda x(l - x)$, find the displacement of the string at any distance x from one end at any time t . [10]

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